MaoMi

Modular anatomy and optics for Microscopy

1. Speed

49.1x faster

0.00667% image difference from MATLAB

4225 propagation sites in both implementations

Original MATLAB CPU: 179.644 s. MaoMi Standard CUDA: 3.662 s.

Same 512 x 512 x 60 tissue, scattering input, pupil and 4225 sites. Full-volume relative L2: 0.00667%. Three serial calls, median; Ryzen 9 7950X / RTX 3080, CUDA batch 8. Optical stage includes phase preparation, propagation, both tails and mask assembly; input generation, static scanning, startup and I/O excluded. Standard-mode comparison, measured 19 October 2026 UTC.

2. Extensibility

Two explicit modes: standard MaoMi preserves MATLAB v4 assembly and native CUDA workflows. RANA (Ray Aware Nonuniform Acquisition) retains each position's full PSF shape and gain with cubic XY interpolation and depth-dependent PSF banks.

RANA: theoretically higher accuracy through four implemented extensions: refractive-index/depth aberration, high-NA vectorial optics, finite-voxel/scan integration and coherent pulse/spectral response.

Area	MaoMi	NAOMi.jl
Execution	Batched CUDA propagation	CPU / FFTW
Native tissue optics	Scattering and multiple excitation modes	Cortical driver deferred; Gaussian example

NAOMi.jl: pinned 17 May 2026 snapshot. Source comparison only; no Julia timing claim.

3. Connections and capabilities

Connection	Available now
RayOptics / wave solvers	Sampled local PSFs, pupil OPD/amplitude callbacks and portable field banks.
RANA physics	Layered vector fields, polarization, voxel/scan dwell, dispersion and spectra.
Cells and paired data	Editable cells, clean/noisy pairs, latent density, seeds and scene splits.

RANA 256 x 256 x 16 control: 0.1030% full-volume difference from higher quadrature; CPU/CUDA rendering 0.206/0.088 s (2.3x). PSF construction: 23.42 s CPU, measured separately.

4. What's next

Extend editable 3-D cells and dynamics, calibrate optical/tissue parameters against real data, and generate paired ground truth for denoising and reconstruction. Add optical providers and acquisition models through the same Python interfaces.

Based on NAOMi: Song et al. (2021), J. Neurosci. Methods 358, 109173.